Milestone 2 Data audit and report

https://github.com/eunangallagher/exoplanet_system_classifier

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I have opened, cleaned, preprocessed and split the dataset into x/y_train and x/y_test
datasets. Then i used a dummy classifier as my baseline model to determine how much better
trained models will perform.

Module Versions:

- Pandas version: 3.0.6
- NumPy version: 2.5.4
- Scikit-learn version: 1.9.1

Data Audit

- Raw data shape: (6451, 10)
- Missing values check:
  pl_name 0
  hostname 0
  default_flag 0
  sy_snum 0
  sy_pnum 0
  st_teff 773
  st_rad 841
  st_mass 877
  st_met 2009
  st_logg 1136
  dtype: int64
- Cleaned Data
  Clean systems left: 3343
  X_train shape: (2674, 6)
  X_test shape: (669, 6)
- Class counts in training:
  0 (Single); 2204
  1 (Multi); 470
- Class percentages in training:
  0 0.824233
  1 0.175767

##Dummy Classifier Performance:

- Overall Baseline Accuracy: 82.36%
- Single-planet Class Accuracy: 100.00%
- Multi-planet Class Accuracy: 0.00%
- Confusion Matrix:
  [[551 0]

[118 0]]

True Positives: 0
False Positives: 0
False Negatives: 118
True Negatives: 551

Dataset choice and target variable setup.

Source: The Planetary Systems Composite Parameters (PSCompPars) table from the official NASA Exoplanet Archive.
Inupt features: 6 physical stellar properties (`number of stars: sy_snum`, `effective temperature: st_teff`, `stellar radius: st_rad`, `stellar mass: st_mass`, `stellar metallicity: st_met`, `surface gravity: st_logg`).
Target variable: I grouped the dataset into each unique stellar system, then i counted the number of planets in each system and set the target variable = 0 if there is one planet in the system and set the target variable = 1 if there is more than 1 planet. The target distribution is heavily skewed to systems with 1 planet.
Train/test split: I chose an 80/20 split to maximise the data my model can train on while keeping enough data to evaluate its performance well. And i stratified by the target variable to retain class proportions in the train and test sets.

Assitance Declaration

I used GitHub CoPilot in vscode as an assitant for the project milestone. I used it to help with pandas and scikit-learn method syntax. All code has been checked and verified by me to ensure it work